1. Definitions and recognition (Sepsis-3)

Definitions

- **Sepsis** = life-threatening organ dysfunction from a dysregulated host response to infection: **SOFA rise ≥ 2**
- **Septic shock** = vasopressor to keep **MAP ≥ 65** + **lactate > 2 mmol/L** despite adequate fluid (mortality > 40%)
- qSOFA (RR ≥ 22, altered mentation, SBP ≤ 100) = prognostic only; **not** a screening tool (SSC 2021)

Exam opening

- ABCDE + "this is septic shock from an abdominal source: time-critical"
- Senior help, ICU, surgeon/IR early; parallel not serial actions

Key point: State the definitions in one line each, then move straight to management.

2. Hour-1 bundle and initial resuscitation

Within 1 hour

- Lactate (repeat if > 2)
- Blood cultures before antibiotics (without delaying them)
- Broad-spectrum antibiotics
- 30 mL/kg crystalloid for hypotension or lactate ≥ 4 (SSC: "suggest", weak)
- Vasopressor if MAP < 65 during or after fluids

Targets

- MAP 65 (65 trial: 60–65 in > 65 yr not worse)
- Lactate clearance or **capillary refill time** (ANDROMEDA-SHOCK: CRT-guided ≥ lactate-guided)
- Urine > 0.5 mL/kg/h, mentation

Key point: Resuscitate to perfusion targets, not to CVP.

3. Fluids: how much and which

How much

- After initial bolus: give only if **fluid responsive** (PLR, PPV/SVV if ventilated, VTI change)
- CLOVERS (2023) & CLASSIC (2022): restrictive = liberal for mortality
- Watch for overload: rising FiO2, B-lines, VExUS congestion

Which

- **Balanced crystalloids** preferred (SMART 2018: ↓ MAKE30; BaSICS & PLUS: no mortality difference)
- Saline: hyperchloraemic acidosis, possible AKI
- Albumin when large volumes needed (ALBIOS: no overall benefit; subgroup in shock)
- **Avoid starches** (6S, CHEST: AKI, RRT, death); gelatins not recommended

Key point: Balanced crystalloid first, assess responsiveness before every further bolus.

4. Source control

Principles

- Identify source early: CT abdomen, US, line review
- Intervene **within 6–12 h** once resuscitated enough
- Least invasive effective option: percutaneous drainage > surgery when possible
- Remove infected devices (lines) after new access

Examples

- Perforated viscus → laparotomy / laparoscopic washout
- Obstructed infected kidney → nephrostomy or stent
- Cholangitis → ERCP; necrotising fasciitis → radical debridement

Key point: Antibiotics cannot sterilise an undrained source: say "source control within 6–12 hours".

5. Antimicrobials

Timing and choice

- Septic shock: **within 1 hour** (Kumar 2006: each hour delay ↑ mortality)
- Cover likely organisms + local resistance + risk factors (recent antibiotics, hospital, ESBL, MRSA, Pseudomonas)
- Abdominal: piperacillin-tazobactam or carbapenem ± anti-fungal if high risk

PK/PD

- Loading dose regardless of renal function; ↑ Vd in sepsis
- Prolonged / continuous β-lactam infusion (BLING III 2024: ↓ mortality signal in meta-analysis)
- TDM for vancomycin, aminoglycosides, β-lactams

Duration

- Short course after source control: STOP-IT (4 days ≈ longer)
- **Procalcitonin** to stop (SAPS, ADAPT-Sepsis): stop when ↓ ≥ 80% or < 0.5 μg/L; never to withhold initial therapy
- Daily de-escalation review

Key point: "Hit hard and early, then de-escalate and stop early."

6. Vasopressors and inotropes

Sequence

- **Noradrenaline** first line; peripheral start acceptable short-term
- Add **vasopressin** 0.03 U/min when NA 0.25–0.5 µg/kg/min (VASST: NA-sparing; possible benefit in less severe shock)
- Adrenaline as second/third agent; angiotensin II (ATHOS-3) in refractory vasoplegia

Cardiac dysfunction

- Echo: septic cardiomyopathy → dobutamine if low output despite MAP
- Levosimendan: no benefit (LeoPARDS)

Steroids

- Hydrocortisone 200 mg/day if ongoing vasopressor (ADRENAL: faster shock reversal; APROCCHSS: ↓ mortality with fludrocortisone)

Key point: Noradrenaline → vasopressin → hydrocortisone → echo-guided inotrope.

7. Emerging concepts and controversies

Metabolic resuscitation

- Vitamin C + thiamine + hydrocortisone: VITAMINS, VICTAS negative; **LOVIT: harm** with vitamin C
- Thiamine only if deficient (alcohol, malnutrition)

Immunomodulation

- Sepsis endotypes (hyper- vs hypo-inflammatory), precision medicine
- IL-1 blockade in macrophage activation-like syndrome; GM-CSF / IFN-γ in immunoparalysis: research only
- Blood purification (CytoSorb, polymyxin B: EUPHRATES negative)

Other

- Beta-blockade (landiolol STRESS-L: no benefit)
- Personalised MAP targets; early vasopressors (CLOVERS)

Key point: Mention promising ideas but conclude "not recommended outside trials".

8. Exam pearls

Structure (10 marks)

- Definition and severity → resuscitation targets → fluids (type/amount, evidence) → source control → antimicrobials (timing, PK, duration, PCT) → vasopressors/steroids → emerging → summary

Numbers to quote

- MAP 65, lactate > 2, 30 mL/kg, 1 hour, 6–12 h source control, NA 0.25 µg/kg/min for vasopressin, hydrocortisone 200 mg/day

Key point: Name a trial for every recommendation: examiners reward evidence.